

Optimizing Decisions for Personal and Business Development

IEDA 1250

Hanyan DUAN

The Hong Kong University of Science and Technology



① Solve LP graphically

② Duality theorem

③ Practice problem

Problem

$$\begin{array}{ll}\text{Maximize} & 3x_1 + 2x_2 \\ \text{s.t.} & x_1 + x_2 \geq 2 \\ & 2x_1 + x_2 \leq 4 \\ & x_1 + 2x_2 \leq 4\end{array}$$

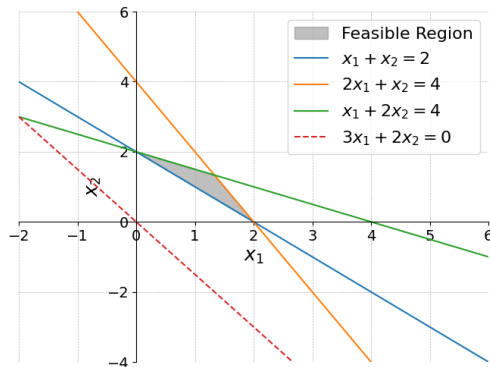
- Objective Function: $3x_1 + 2x_2$
- Decisions variables: x_1, x_2
- A LP maximizes/minimizes a linear function subject to a set of constraints given by linear inequalities/equalities.

Problem

$$\begin{array}{ll}\text{Maximize} & 3x_1 + 2x_2 \\ \text{s.t.} & x_1 + x_2 \geq 2 \\ & 2x_1 + x_2 \leq 4 \\ & x_1 + 2x_2 \leq 4\end{array}$$

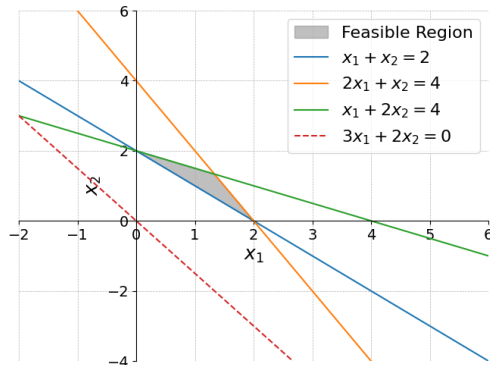
Feasible.

$$z = 3x_1 + 2x_2 \longrightarrow x_2 = \frac{1}{2}z - \frac{3}{2}x_1$$



Problem

$$\begin{array}{ll}\text{Maximize} & 3x_1 + 2x_2 \\ \text{s.t.} & x_1 + x_2 \geq 2 \\ & 2x_1 + x_2 \leq 4 \\ & x_1 + 2x_2 \leq 4\end{array}$$

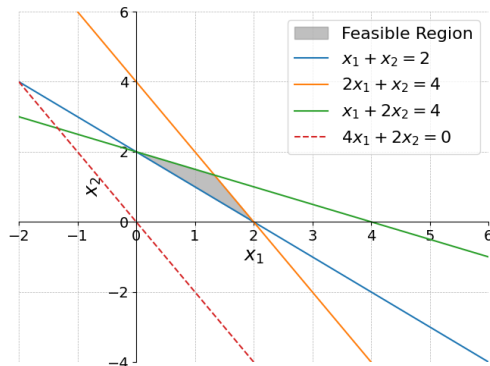


$$z = 3x_1 + 2x_2 \longrightarrow x_2 = \frac{1}{2}z - \frac{3}{2}x_1$$

One optimal solution.

Problem

$$\begin{array}{ll}\text{Maximize} & 4x_1 + 2x_2 \\ \text{s.t.} & x_1 + x_2 \geq 2 \\ & 2x_1 + x_2 \leq 4 \\ & x_1 + 2x_2 \leq 4\end{array}$$



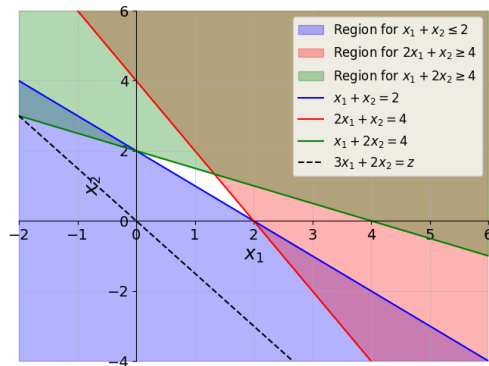
$$z = 4x_1 + 2x_2 \longrightarrow x_2 = \frac{1}{2}z - 2x_1$$

Feasible. Multiple solutions.

Problem

$$\begin{array}{ll}\text{Maximize} & 3x_1 + 2x_2 \\ \text{s.t.} & x_1 + x_2 \leq 2 \\ & 2x_1 + x_2 \geq 4 \\ & x_1 + 2x_2 \geq 4\end{array}$$

Infeasible.

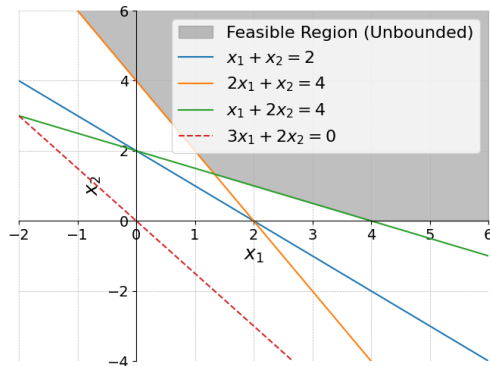


Problem

$$\begin{array}{ll}\text{Maximize} & 3x_1 + 2x_2 \\ \text{s.t.} & x_1 + x_2 \geq 2 \\ & 2x_1 + x_2 \geq 4 \\ & x_1 + 2x_2 \geq 4\end{array}$$

$$z = 3x_1 + 2x_2 \longrightarrow x_2 = \frac{1}{2}z - \frac{3}{2}x_1$$

Unbounded.



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- Two sets: P and Q
- Three kinds of fruit: apple(A), banana(B), and orange(C)
- Objective: maximum profit
- Constraints: available inventory

	A	B	C
P (45/unit)	20	12	15
Q (60/unit)	10	28	6
Inventory	1800	1440	1440

$$\begin{array}{ll} \text{Maximize} & Z = 45x_1 + 60x_2 \\ \text{s.t.} & 20x_1 + 10x_2 \leq 1800 \quad (A) \\ & 12x_1 + 28x_2 \leq 1440 \quad (B) \\ & 15x_1 + 6x_2 \leq 1440 \quad (C) \\ & x_1, x_2 \geq 0 \end{array}$$

Primal

$$\begin{array}{ll} \text{Maximize} & Z = 45x_1 + 60x_2 \\ \text{s.t.} & 20x_1 + 10x_2 \leq 1800 \quad (A) \\ & 12x_1 + 28x_2 \leq 1440 \quad (B) \\ & 15x_1 + 6x_2 \leq 1440 \quad (C) \\ & x_1, x_2 \geq 0 \end{array}$$

Dual

$$\begin{array}{ll} \text{Minimize} & W = 1800y_1 + 1440y_2 + 1440y_3 \\ \text{s.t.} & 20y_1 + 12y_2 + 15y_3 \geq 45 \quad (P) \\ & 10y_1 + 28y_2 + 6y_3 \geq 60 \quad (Q) \\ & y_1, y_2, y_3 \geq 0 \end{array}$$

Primal

$$\begin{array}{ll} \text{Maximize} & Z = 6x_1 + x_2 + x_3 \\ \text{s.t.} & 4x_1 + 3x_2 - 2x_3 = 1 \\ & 6x_1 - 2x_2 + 9x_3 \geq 9 \\ & 2x_1 + 3x_2 + 8x_3 \leq 5 \\ & x_1 \geq 0, x_2 \leq 0, x_3 \text{ unrestricted} \end{array}$$

Primal

$$\begin{array}{ll} \text{Maximize} & Z = 6x_1 + x_2 + x_3 \\ \text{s.t.} & 4x_1 + 3x_2 - 2x_3 = 1 \\ & 6x_1 - 2x_2 + 9x_3 \geq 9 \\ & 2x_1 + 3x_2 + 8x_3 \leq 5 \\ & x_1 \geq 0, x_2 \leq 0, x_3 \text{ unrestricted} \end{array}$$

Dual

$$\begin{array}{ll} \text{Minimize} & W = y_1 + 9y_2 + 5y_3 \\ \text{s.t.} & 4y_1 + 6y_2 + 2y_3 \geq 6 \\ & 3y_1 - 2y_2 + 3y_3 \leq 1 \\ & -2y_1 + 9y_2 + 8y_3 = 1 \\ & y_1 \text{ unrestricted}, y_2 \leq 0, y_3 \geq 0 \end{array}$$

Primal model (or dual model)	Dual model (or primal model)
(MAX)	(MIN)
constraint j is $\leq$	variable $y_j \geq 0$
constraint j is $=$	variable y_j is unrestricted
constraint j is $\geq$	variable $y_j \leq 0$
variable $x_i \geq 0$	constraint i is $\geq$
variable x_i is unrestricted	constraint i is $=$
variable $x_i \leq 0$	constraint i is $\leq$

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Practice Example

Failsafe Electronics Corporation primarily manufactures four highly technical products, which it supplies to aerospace firms that hold NASA contracts. Each of the products must pass through the following departments before they are shipped: wiring, drilling, assembly, and inspection. The time requirements in each department (in hours) for each unit produced and its corresponding profit value are summarized in this table. What is the optimal Product mix?

Product	Wiring	Drilling	Assembly	Inspection	Unit Profit
XJ201	.5	3	2	.5	\$9
XM897	1.5	1	4	1.0	\$12
TR29	1.5	2	1	.5	\$15
BR788	1.0	3	2	.5	\$11
Available hours	1500	2350	2600	1200	

Maximize $Z = 9x_1 + 12x_2 + 15x_3 + 11x_4$ (Total Profit)

subject to

$$0.5x_1 + 1.5x_2 + 1.5x_3 + 1.0x_4 \leq 1500 \quad (\text{Wiring})$$

$$3.0x_1 + 1.0x_2 + 2.0x_3 + 3.0x_4 \leq 2350 \quad (\text{Drilling})$$

$$2.0x_1 + 4.0x_2 + 1.0x_3 + 2.0x_4 \leq 2600 \quad (\text{Assembly})$$

$$0.5x_1 + 1.0x_2 + 0.5x_3 + 0.5x_4 \leq 1200 \quad (\text{Inspection})$$

$$x_1, x_2, x_3, x_4 \geq 0 \quad (\text{Non-negativity})$$

$$x_1, x_2, x_3, x_4 \in \mathbb{Z} \quad (\text{Integer constraints})$$

Thank you for listening !